

Lucas B. Filla

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EDUCATION

Ohio University, Russ College of Engineering | Athens, OH

Anticipated Graduation: May 2028

Bachelor of Science

GPA: 4.0

Majors: Computer Science, Artificial Intelligence

Minor: Business Administration

EXPERIENCE

Ohio University OIT Department | Athens, OH

August 2025 – Present

Student Software Engineer

- Modernized legacy Angular and Node.js applications within an Agile team, writing technical documentation and managing the project's lifecycle from planning through deployment.
- Engineered a task manager application with robust features utilizing Angular, Node.js, TypeScript, and SQL, while implementing comprehensive unit testing, validation, and HTTP routing protocols.
- Utilized Docker development containers to streamline development across devices, eliminating environment inconsistencies across the engineering cohort.

The Ohio State University AI-EDGE REU | Columbus, OH

June 2026 – July 2026

Undergraduate Research Fellow

- Conceptualized computer vision pipelines for NSF-funded Project SCARLET, collaborating with OSU Athletics and Google Engineering to measure athlete biomechanics from LiDAR and MediaPipe.
- Developed automated static anthropometric and functional mobility assessments utilizing ARKit, Vertex AI, and NumPy, targeting ± 1 cm measurement accuracy without external lab equipment.
- Designed an intuitive, Swift-based, native iOS interface for on-device static and kinetic clinical assessments, ensuring streamlined data collection by any user.

PROJECTS

SylvaSense NRT | *Python, PyTorch, Google Earth Engine API, Rasterio, NumPy*

- Architected an automated data pipeline using the Google Earth Engine API to extract, synchronize, and normalize multi-spectral Sentinel-2 satellite imagery with ground-truth deforestation datasets.
- Developed a custom U-Net deep learning architecture in PyTorch, implementing skip connections to accurately classify near real-time deforestation patterns across the Amazon.
- Processed multi-gigabyte spatial tensors using sliding window algorithms, optimizing raw 32-bit float GeoTIFFs into localized, memory-efficient arrays to prevent GPU saturation during model training.

SnipBoard | *Qt, QML, SQLite, C++*

- Designed the frontend for a cross-platform desktop application using the Qt framework and C++ to help developers efficiently store, organize, and retrieve reusable code snippets.
- Collaborated within a 4-person agile team to integrate the frontend architecture seamlessly with a custom C++ backend, local SQLite database, and a native VSCode extension.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL, HTML/CSS, Swift

Frameworks: Angular, Node.js, PyTorch, NumPy, Vitest, MediaPipe, ARKit, Rasterio

Tools: Git, GitHub, Docker, Jira, Kubernetes, Postman, Google Earth Engine, Figma